

Calculus II: Sequences and Series Survival Sheet

Sequences

Definition 1.

A *sequence* is a function $a_n : \mathbb{N} \rightarrow \mathbb{R}$.

$$\{a_n\} = a_1, a_2, a_3, \dots$$

This is an infinitely long ordered list.

Limit of a Sequence

Definition 2.

The sequence $\{a_n\}$ has the *limit* L and we write

$$\lim_{n \rightarrow \infty} a_n = L.$$

If $\lim_{n \rightarrow \infty} a_n$ exists, we say the sequence *converges*. Otherwise, we say the sequence *diverges*.

Theorem 0.1. *If*

$$\lim_{n \rightarrow \infty} f(x) = L.$$

and $f(n) = a_n$ where n is an integer, then $\lim_{n \rightarrow \infty} a_n = L$.

Example: $1, -1, 1, -1, \dots$ is of the form $(-1)^{n+1}$

Example: $-1, 1, -1, 1, \dots$ is of the form $(-1)^n$

Algebraic Limit Laws

If $\lim a_n = A$ and $\lim b_n = B$, then:

$$\lim(a_n + b_n) = A + B$$

$$\lim(a_n b_n) = AB$$

$$\lim \frac{a_n}{b_n} = \frac{A}{B}, \quad B \neq 0$$

Squeeze Theorem

If

$$a_n \leq b_n \leq c_n$$

and

$$\lim a_n = \lim c_n = L,$$

then

$$\lim b_n = L.$$

Infinite Series

Definition 3.

An infinite series is

$$\sum_{n=1}^{\infty} a_n.$$

Definition 4.

Define partial sums:

$$S_N = \sum_{n=1}^N a_n.$$

Example: These are partial sums:

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

In general we have:

$$s_n = a_1 + a_2 + \dots + a_n$$

The series converges if $\lim_{N \rightarrow \infty} S_N$ exists.

Definition 5.

Given a series $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots$, let s_n denote its n th partial sum.:

$$s_n = \sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \dots + a_n.$$

If the sequence $\{s_n\}$ is convergent and $\lim_{n \rightarrow \infty} s_n = s$ exists as a real number, then the series $\sum_{n=1}^{\infty} a_n$ is called *convergent* and we write:

$$a_1 + a_2 + a_3 + \dots + a_n = s.$$

or

$$\sum_{i=1}^{\infty} a_n = s$$

The number s is called the *sum* of the series. If the sequence $\{s_n\}$ is divergent, the the series is called *divergent*.

Divergence Test

If

$$\lim_{n \rightarrow \infty} a_n \neq 0,$$

then the series diverges. If it is 0, no conclusion can be drawn.

Note: This test can be used on any series.

Linearity of Series

$$\sum_{n=1}^{\infty} c a_n = c \sum_{n=1}^{\infty} a_n, \quad c \in \mathbb{R}$$
$$\sum_{n=1}^{\infty} (a_n \pm b_n) = \sum_{n=1}^{\infty} a_n \pm \sum_{n=1}^{\infty} b_n$$

Geometric Series

A *geometric series* is of the form:

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots$$

where $a \neq 0$

Each term is obtained from the preceding one by multiplying it by the *common ratio* (r).

The geometric series

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots$$

is convergent if $|r| < 1$ and its sum is:

$$\frac{a}{1-r}.$$

If $|r| \geq 1$ the geometric series is divergent.

Note: This is specifically for the geometric series.

p-Series

A *p-series* is of the form:

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

- Converges if $p > 1$
- Diverges if $p \leq 1$

Harmonic Series:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

Using the p-series test: $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$.
Here, $p = 1$, so the series diverges.

Integral Test

Let f be continuous from $[1, \infty)$, positive, decreasing and $a_n = f(n)$.

Then

$\sum a_n$ converges if and only if $\int_1^\infty f(x) dx$ converges

$\sum a_n$ diverges if and only if $\int_1^\infty f(x) dx$ diverges

either both converge or both diverge.

Comparison Tests

Direct Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

- If $\sum b_n$ converges and $a_n \leq b_n \forall n \Rightarrow \sum a_n$ converges.
- If $\sum b_n$ diverges and $a_n \geq b_n \forall n \Rightarrow \sum a_n$ diverges.

Limit Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c, \quad 0 < c < \infty,$$

then either both series converge or both diverge.

Alternating Series

Definition 6.

An *alternating series* is a series whose terms are alternately positive and negative.

Alternating Series Test

For

$$\sum (-1)^n b_n,$$

if:

- $b_{n+1} \leq b_n \forall n$,
- $\lim_{n \rightarrow \infty} b_n = 0$,

then the series converges.

Note: b_n is the absolute value of a_n

Absolute and Conditional Convergence

Definition 7.

A series is *absolutely convergent* if $\sum |a_n|$ converges.

Definition 8.

A series is *conditionally convergent* if it is convergent but not absolutely convergent ($\sum |a_n|$ diverges).

Theorem 0.2. *If a series $\sum a_n$ is absolutely convergent, then it is convergent.*

Ratio Test

Let

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

- If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$ the series $\sum a_n$ is absolutely convergent (So it is convergent)
- If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L > 1$ the series $\sum a_n$ is divergent
- If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L = 1$ then the ratio test is inconclusive for the series $\sum a_n$.

Power Series

$$\sum_{n=0}^{\infty} c_n (x - a)^n$$

Radius of Convergence

Find R using Ratio Test.

- Converges for $|x - a| < R$
- Diverges for $|x - a| > R$
- Test endpoints separately

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Maclaurin Series (Taylor Series with $a = 0$)

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}, \quad |x| < 1$$